HARAMAYA UNIVERSITY

COLLEGE OF COMPUTING AND INFORMATICS

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INTODUCTION

The AI-powered tutoring system project is designed to improve the learning experience of students through affordable, personalized tutoring support. The system will leverage cutting-edge technologies such as AI-powered chatbots to deliver immediate, contextually relevant answers to student queries, and integrate interactive features such as quizzes and progress monitoring to enhance student engagement and learning achievements. The platform will be easy to use, with an intuitive interface for both students and tutors.

The process of development will be organized into a number of different phases, each of which makes a contribution to the overall functionality and effectiveness of the system. These are requirements collection, system design, content creation, integration of AI, user interface creation, tracking of progress, and testing. The system will also be rolled out and tested in actual educational environments to ensure usability, effectiveness, and scalability. The ultimate aim is to provide an instrument that has the potential to assist students with learning difficulties and give tutors an effective platform through which they can offer assistance.

This project will be undertaken within the period of 14 weeks, with three people contributing in research, system design, development, content generation, and testing. Each member will have particular roles to attend to, and all project areas will be addressed accordingly. The success of the project will be determined by how usable the system is, how it affects learning results, through positive comments from the users, and how long the users continue using it.

1. Team Profile

- **Tewodros Fikadu**: A visionary in artificial intelligence, Tewodros excels at crafting adaptive systems that learn from user interactions, ensuring the tutoring platform delivers highly personalized learning experiences that feel like one-on-one coaching. His ability to translate complex data patterns into tailored educational content is unmatched, making him the driving force behind the system's intelligence.
- **Tesfaye Eshetu**: A creative genius in web design, Tesfaye specializes in building visually stunning and intuitive interfaces that captivate users. His talent for creating seamless, engaging platforms ensures that students and tutors find the system effortless to use, turning learning into an immersive and enjoyable journey.
- **Tolesa Tesfaye**: A master of organization and communication, Tolesa is a project management dynamo who keeps the team on track with laser focus. Her exceptional documentation skills and knack for presenting ideas clearly ensure that every milestone is met with precision, making the project's progress smooth and transparent.
- **Worku Teshome**: A data wizard with a passion for efficiency, Worku thrives at managing vast amounts of information, ensuring the system organizes and delivers educational content flawlessly. His expertise in handling complex datasets guarantees that student progress and learning materials are always accessible and perfectly structured.

- **Team Leader**: Tewodros Fikadu, who serves as an inspiring facilitator, orchestrating team meetings, aligning efforts, and keeping the project's vision in sharp focus. His leadership fosters collaboration and drives the team toward delivering an extraordinary tutoring system.
- **Team Strengths and Size Justification**: This powerhouse team combines unparalleled expertise in artificial intelligence, web design, project management, and data organization, creating a perfect synergy for building an intelligent, user-friendly tutoring platform. Tewodros's AI prowess ensures personalized learning, Tesfaye's design finesse guarantees an engaging interface, Tolesa's organizational brilliance keeps the project on track, and Worku's data mastery ensures seamless content delivery. A four-member team is essential, as the project demands simultaneous development of sophisticated components: adaptive

learning paths, an intuitive interface, robust data systems, and meticulous project coordination. A smaller team would risk delays or quality trade-offs, unable to handle the workload within the 3-week timeline. This dynamic quartet ensures each member shines in their specialty while collaborating on testing and refinement, delivering a robust, game-changing solution.

2. Problem diagnosis

❖ Problem Domain and Existing Challenges

The problem domain is e-learning and education, specifically the delivery of personalized academic support and the tutoring of students through electronic modes. Since flexible and autonomous learning is increasingly in demand, institutions have shifted to using e-learning systems. However, such systems are typically deficient when it comes to responding to individual students' needs in a dynamic and interactive way.

In current practice, students typically access static online learning materials such as lecture notes, pre-recorded videos, or static quizzes. These are unchanging based on student performance and learning trends. Moreover, exposure to live tutoring is limited as a result of constraints such as tutor availability, cost, and scheduling. It leaves one solely dependent on independent study, potentially ineffective for the student who needs intense support or is not even motivated.

❖ There are several specific issues that have been raised:

- ✓ **No individualization:** All students receive the same content irrespective of their attainment, study rate, or knowledge holes.
- ✓ **No real time support:** Students struggle and need to wait for feedback, which disrupts the learning flow.
- ✓ **Useless monitoring of progress:** Current systems fail to provide meaningful indications of where the student is faltering, nor do they suggest remedial interventions.
- ✓ **Tutor overloading:** Human tutors cannot spend individual time for every student, especially in large classes, thus learning is missed.

- ✓ **Low involvement**: Without adaptive or interactive content, many students are bored and disinterested with online learning contexts.

3. Proposed Treatment

How the Proposed System Will Solve the Diagnosed Problems As a remedy to the issues identified in the current tutoring and learning practices, we propose a web-based AI-powered tutoring system that delivers personalized, interactive, and real-time support to university students. The system will leverage artificial intelligence to dynamically assess the performance and learning style of each student and adapt the learning experience accordingly. This approach simulates human tutoring practice while scaling the support to a very large number of students.

3.1 Goal of AI-powered Tutoring

- ➢ The main goals of the proposed system are:

- ✓ **Provide On-Demand Academic Support:** Offer 24/7 access to a virtual tutor so students can get help at any time, regardless of location or availability of human tutors.
- ✓ **Enhance Learning Outcomes:** Use interactive lessons, quizzes, and personalized feedback to help students understand difficult topics and improve performance.
- ✓ **Encourage Independent Learning:** Empower students to study at their own pace, revisit concepts as needed, and build confidence without fear of judgment.
- ✓ **Reduce the Workload of Human Tutors:** Automate routine tasks like answering common questions and checking quizzes, allowing tutors to focus on more complex student needs.
- ✓ **Monitor and Visualize Progress**: Help students and tutors track learning achievements and identify areas where students need more support through dashboards and analytics.
- ✓ **Promote Equitable Access to Help:** Ensure all students, regardless of background or schedule, have equal access to academic assistance.
- ✓ **Foster Feedback-Driven Improvement:** Collect user feedback continuously to improve content, usability, and learning impact.
- ✓ **Support Academic Departments:** Provide a useful tool for departments to improve course support and reduce dropout rates in challenging subjects.

❖ Targeted Interventions and Achieving the Objectives

➢ Targeted Interventions of the System: The proposed AI-powered tutoring system incorporates the following key interventions to improve academic outcomes:

- ✓ **AI-Driven Personalization:** Monitors student performance (e.g., quiz scores, time spent on topics, interaction patterns) to generate individualized learning paths and practice exercises tailored to each student's needs.
- ✓ **Real-Time Q&A Chatbot:** An integrated AI chatbot provides immediate responses to academic queries, reducing reliance on the limited availability of human tutors.
- ✓ **Immediate Feedback Engine:** Offers instant explanations and actionable feedback whenever students complete quizzes or make mistakes, enhancing their understanding and learning retention.
- ✓ **Progress Analytics:** Visual dashboards track student performance over time, highlighting strengths, weaknesses, time investment, and learning progress to both students and tutors.
- ✓ **Adaptive Content Recommendation:** Based on performance data, the system suggests additional videos, readings, or practice materials to reinforce concepts

❖ Dynamic Learning Pathways and Scalability

- ✓ **Evolving Learning Paths:** The system adapts dynamically as the student progresses, ensuring that content, difficulty, and support match their current understanding and pace of learning.
- ✓ **Reduced Tutor Workload:** Routine support tasks like answering common questions and evaluating standard quizzes are automated, allowing tutors to focus on strategic, high-impact academic mentoring.
- ✓ **High Scalability:** The AI-driven platform can support hundreds of learners simultaneously without compromising the quality of feedback or personalization, making it ideal for large university classes.

3.2 Services Provided by the System:

The AI-Powered Tutoring Web-Based Application provides the following critical services to facilitate learning and instructional guidance:

- **AI-Powered Q&A Chatbot:** Allows students to ask academic questions in their own words. Presents immediate, clear, and correct answers via AI. Reduces response wait times and replicates a live tutoring experience.
- **Interactive Quizzes with Feedback:** Offers quizzes on specific subjects or classes. Offers instant feedback, complete with wrong answer explanations. Aids students to assess and improve their knowledge.
- **Personalized Learning Dashboard:** Displays each student's learning journey, quiz results, and suggested topics. Adapts based on performance and learning trend. Encourages goal setting and self-directed learning.
- **Tutor Management Interface:** Enables tutors to upload learning materials and quizzes. Enables tutors to view student progress, identify struggling students, and intervene early. Streamlines content delivery and monitoring tasks.
- **User Authentication and Access Control:** Ensures secure access to the system through login and role-based authorization. Protects personal and academic data. Maintains a clean distinction between student and tutor access.
- **Educational Content Repository:** Holds learning materials such as videos, notes, examples, and practice questions. Collected by subject, topic, and difficulty level for quick access. Enables continuous learning outside of class.
- **Analytics and Progress Tracking:** Tracks and shows student activity, topic mastery, and performance trends data. Helps both students and tutors identify strengths and weaknesses. Enables data-driven academic decisions and personalized advice.

Summary of Functional Features

- **For Students:**

✓ Log in to a personalized learning dashboard.

- ✓ Ask questions using an AI-powered chatbot.
- ✓ Take interactive quizzes and receive instant feedback.
- ✓ Track learning progress with visual summaries.
- ✓ Access and review educational content and examples.

➢ **For Tutors:**

- ✓ Upload and organize new content and quizzes.
- ✓ View analytics dashboards showing student engagement and progress.
- ✓ Identify struggling students and provide targeted support.
- ✓ Receive feedback from students to help improve learning support.

3.3 Typical Customers and Users

The proposed system is designed primarily for use within university environments. Key user groups include:

Primary Users – Students:
University students, especially those in STEM (Science, Technology, Engineering, and Mathematics) programs, who require flexible and accessible academic support outside of regular classroom hours.

Secondary Users – Tutors and Instructors:
Academic staff responsible for mentoring students, monitoring their academic progress, and providing timely interventions.

Institutional Stakeholders – Academic Departments and Administrators:
University departments focused on improving student learning outcomes, reducing failure and dropout rates, and enhancing overall academic support through data-driven tools.

Note: While no formal agreement has been established, several tutors and students consulted during the research phase have shown strong interest in the system and have volunteered to participate in testing during the development phase.

3.4 Evaluation Metrics for Success

Evaluation Metrics for Success To quantify whether the suggested system is a success, we will use both quantitative and qualitative metrics:

- **Quantitative Metrics**:
  - ✓ **Engagement Rate**: Number of logins and time spent on the platform per user per week.
  - ✓ **Test Performance Improvement**: Percentage increase in post-use test scores compared to pre-use diagnostics.
  - ✓ **Query Response Time**: Average time for the AI tutor to provide useful answers (target: under 5 seconds).
  - ✓ **Learning Module Completion Rate**: Percentage of assigned modules completed, reflecting usability and persistence.

- **Qualitative Metrics**:
  - ✓ **User Satisfaction**: Survey responses rating ease of use, personalization, and helpfulness.
  - ✓ **Perceived Usefulness**: Feedback from student and tutor interviews post-use.
  - ✓ **Adoption Rate**: Number of active users (students and tutors) over time.

3.5 Business Value and Benefits

The proposed system offers considerable business value for the universities and tutors, as well as real benefits for the students:

- ✓ **For Students:** On-demand academic support whenever required, without waiting due to limited tutor availability.Personalized learning experience, with options to focus on areas of weakness.Enhanced motivation and persistence through interactive learning and progress tracking.Academic enhancement as students receive personalized exercises and feedback.
- ✓ **For Tutors and Academics:** Efficient monitoring of the students through analytics and dashboards. Reduced repetitive workload, allowing tutors to focus on more detailed mentoring. Data-driven insights for early detection of struggling students.

Example Scenarios: A student learning for a midterm in Calculus logs in during the evening and is struggling with integration by parts. Instead of waiting for morning office hours, they ask the AI tutor, which explains the steps interactively and provides a practice problem.

A tutor overseeing 80 students in a first-year computer programming course looks at the dashboard and, at a glance, identifies 10 students struggling with recursion, allowing for focused intervention before the next exam.

- ➢ Why It's Better Than Current Practice Current e-learning systems are static, slow to respond, and don't adapt to individual needs. Human tutors are swamped and can't offer 24/7 support. Our proposed system closes this gap by delivering:24/7 tutoring assistance via AI, with real-time and contextual feedback.

4. Plan of Work

To ensure the successful development and deployment of the proposed AI-powered tutoring system, our work is divided into manageable, logical phases. Each phase builds upon the previous one or can be developed independently in parallel by different team members. The estimated duration of the project is 3 weeks, with key milestones as follows:

Phase 1: Requirements Gathering & Research

This phase involves interviewing students and tutors, reviewing existing tutoring methods, and defining user needs and goals. The time estimate for this phase is 2 weeks and it does not depend on any other phases.

Phase 2: System Design

In this phase, the system layout and screen flows will be planned, and core features like the dashboard, chatbot, and quizzes will be defined in non-technical terms. This phase also lasts 2 weeks and depends on the completion of Phase 1.

Phase 3: Content Development

This phase involves creating educational content, such as quizzes, explanations, and feedback templates. This can be done in parallel with Phase 1 and will take 3 weeks.

Phase 4: User Interface Development

In this phase, the development team will build web pages and interactions, including login, dashboard, chatbot, and quiz pages.It will take 2 weeks and depends on the completion of Phase 2.

Phase 5: Progress Tracking & Analytics Module

This phase focuses on designing dashboards for students and tutors to monitor performance. It will take 2 weeks and depends on the completion of Phases 2 and 4.

Phase 6: Testing & Feedback

During this phase, user testing will be conducted with real students and tutors. Feedback will be gathered and used to iterate on the design. This phase will take 2 weeks and depends on the completion of Phases 3–5.

Phase 7: Final Deployment & Training

This final phase involves launching the system and training tutors and admins on how to use the dashboard. It will take 1 week and depends on the successful completion of all previous phases.

➢ Team Description

Our team consists of three members, each contributing complementary strengths:

Member 1: Project Lead & Researcher

Strengths: Excellent communication skills, strong understanding of educational needs, experience in gathering user feedback.

Role: Coordinate meetings with students and tutors, document requirements, oversee testing and user feedback.

Member 2: Developer & System Designer

Strengths: Proficient in web technologies, UI/UX design, experience integrating third-party APIs like OpenAI or Google Dialogflow.

Role: Build the platform, ensure a smooth and intuitive user interface, and integrate AI functionalities.

Member 3: Content Creator & Tester

Strengths: Subject knowledge in computer science and math, strong writing and editing skills, good eye for detail.

Role: Create educational materials, quizzes, and explanations; manage quality control and coordinate user testing.

Why this team size is appropriate:

The scope of the project is broad (education, AI integration, analytics, and web interface), which requires a multi-skill team. A smaller team would be overwhelmed, while a larger team would not be necessary for a prototype-stage solution. This size allows us to work efficiently in parallel.

Measuring SuccessTo know we have succeeded, we will look for the following results:

Usability: Students can navigate the platform and get useful, timely responses from the AI assistant.

Learning impact: At least 70% of users demonstrate improved understanding (measured through before/after quizzes).

Positive feedback: More than 80% of surveyed users (students and tutors) report that the system is helpful, easy to use, and better than current support systems.

Sustained engagement: At least 50% of users return to the system multiple times per week, showing trust and interest. These results will be measured through questionnaires, usage statistics, and user interviews.

❖ User Experience

The system will feel like a helpful online tutor that students can access anytime. After logging in, the student sees a clean dashboard showing their recent topics, performance, and learning recommendations. If they get stuck on a topic like "recursion in programming" or "integration in calculus," they can type a question into the AI assistant, which replies instantly in simple, clear language. If they make mistakes in practice questions, the system shows what went wrong and explains the concept.

Tutors will have access to a backend view where they can upload new materials and see which students need extra help. It saves tutors from answering repetitive questions and lets them focus on students who are really struggling.

Resources and Access Needs

To build and test this system effectively, we need access to:Existing course materials (lecture notes, sample assessments) from the university's computer science and math departments. Students and tutors willing to participate in interviews and testing. Internet access and hosting for deploying the web application (can be done using university resources or free-tier cloud hosting). An AI API service (such as OpenAI's GPT model) for natural language responses.

Conclusion

the AI-based tutoring system is an exceptional innovation in the provision of customized, scalable, and efficient learning support to students. With the integration of AI-powered technologies like a chatbot and interactive quizzes, the system aims to enhance student engagement, facilitate real-time learning, and provide tutors with convenient methods of monitoring progress. The disciplined stages of the project, from requirements gathering through ultimate deployment, guarantee that each element is painstakingly developed and tested to serve the requirements of both tutors and students.

As the system moves forward in its different stages, the team will closely collaborate to develop a system that is intuitive, stable, and effective. Success will be measured not only by the technical functionality of the system but also by its ability to improve learning outcomes, user satisfaction, and engagement. This project has the potential to revolutionize tutoring by offering an innovative, AI-powered solution that helps bridge learning gaps, providing students with valuable, on-demand support while also easing the burden on tutors.